COPADO | QUALITY GATES BEST PRACTICES

KB0099390 - Latest Version

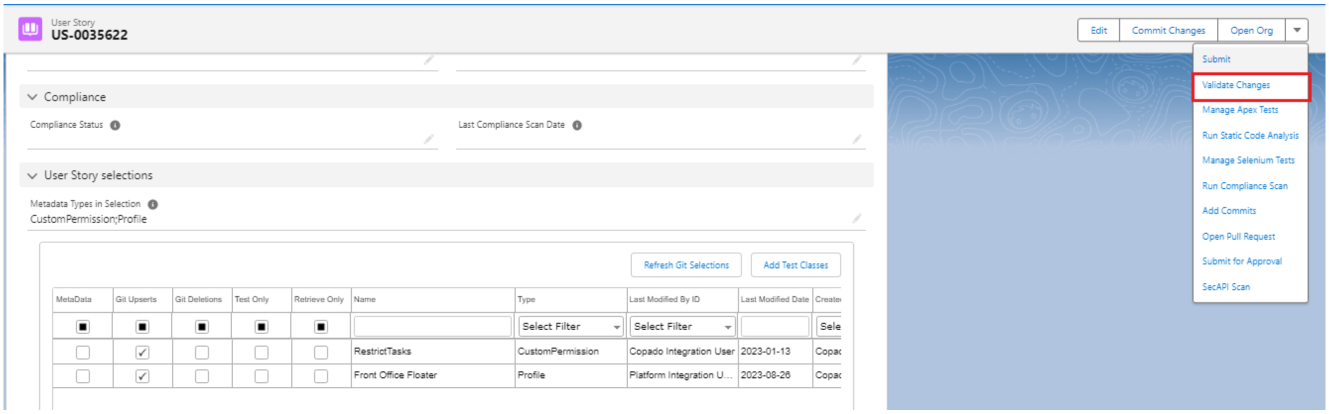
REFERENCE: KB0099390 (https://humanaprod.service-now.com/asc?id=kb_article&sysparm_article=KB0099390)

QUALITY GATES BEST PRACTICES IN COPADO
1. To validate a deployment, package the changes attached to a user story and use Copado to attempt a deployment without actually deploying it, checking for "Completed Successfully" or "Completed with Errors" status.
2. When making changes to an apex class, trigger, or test class, ensure that each apex class has a corresponding test class, either added as metadata or as a test class, and run the apex tests to achieve the required code coverage and passing methods.
3. Utilize Copado's in-house solution to handle pull requests, utilizing a Salesforce Approval Process for PR approval and updating the "Pull Requests Approved" checkbox on the user story record to enable promotion.
4. Use Copado's built-in PMD Code Scan for static code analysis of Apex code, ensuring that all user stories are scanned against the customized ruleset, with the Static Code Analysis Passed checkbox enabling deployment if there are no Priority 1 violations.
5. Prior to Production release, perform a static Checkmarx scan (Security SAST scan) to ensure code is free of security violations that could compromise data.
6. Use the Back Promotion Checker to check for pending back promotions in the target environment and block promotions if necessary.

1. Validation Deployment

A validation deployment packages the changes attached to a user story and attempts (without actually deploying) to deploy the package to the next environment in the pipeline. Copado will provide a status once the package has been run: "Completed Successfully" or "Completed with Errors"

- Commit changes to the user story
- Verify your components are attached to the user story by looking at the metadata grid (screenshot below).
- Click the button "Validate changes"

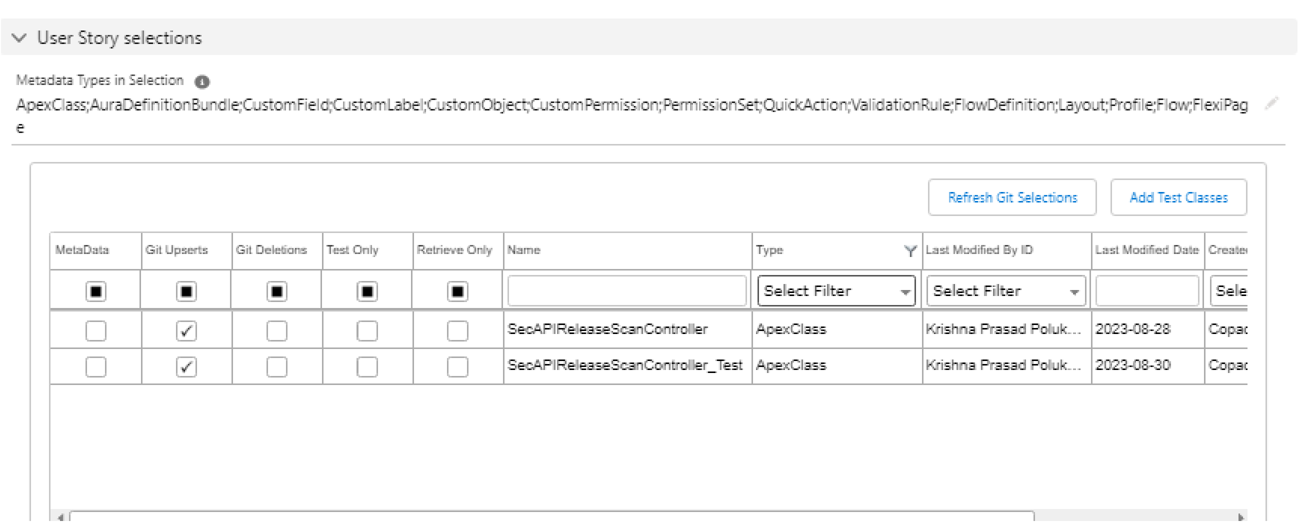


- "Completed Successfully" = ready to deploy your changes forward.
- "Completed with Errors"
  - Click on the last validation deployment name from the user story. This will take you to the validation deployment record.
  - Scroll down to the Steps section. You should see one Git Promotion Step
  - Click on the Red Flag. This will allow you to see the deployment errors.

2. Running Apex Tests on User Story (required only if you've changed an apex class, trigger, or test class)

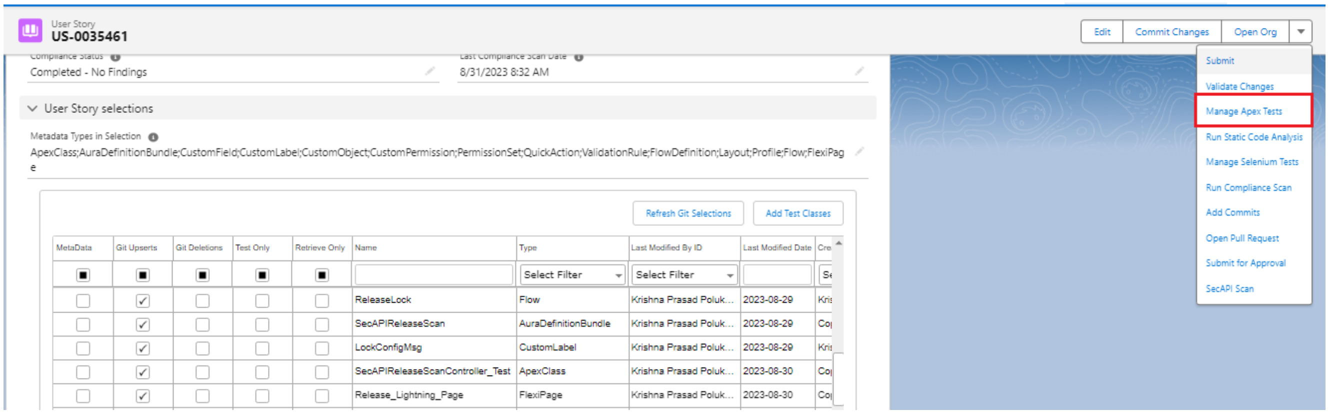
Each apex class should have a corresponding test class. Test classes can be added to a user story in two different ways:

a. Added as metadata. This will be done when the apex class or trigger was created or modified and will be done during the "How do I attach my changes to a User Story"
b. Added as a test class. This feature will be used when you change your apex class and want to make sure the class is still covered by the existing test class that was created for it.

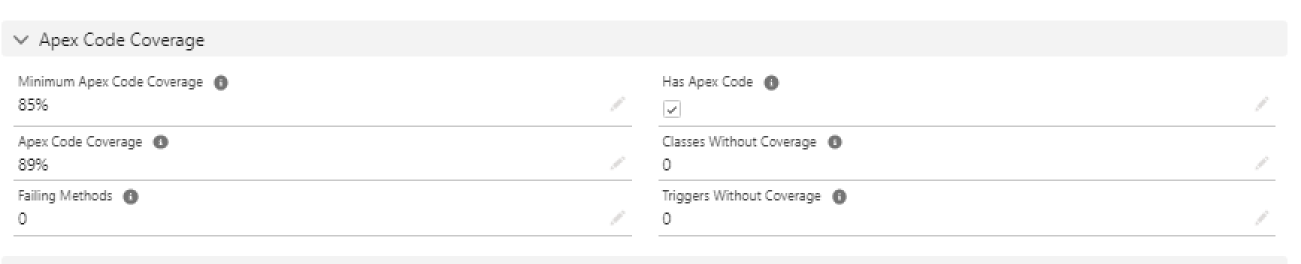


- If the test apex class was created or modified, you will find it attached here after you "committed" the changes to GIT.

- If the test class was not added during the commit process, you should click 'Add Test Classes" and attach the test classes that you need for the apex classes.
- Once the test classes are added through the "Add Test Classes" button, you will see a line item for the test class with a checkbox in the "Test Only" column.
- Run the apex tests.
- Click on Manage Apex Tests



- Click Run Apex Tests
- Once finished, you will be able to review the methods and code coverage for the test classes.
    - 85% code coverage and 100% passing methods is required by Salesforce.
- Test results will be shown on the user story after tests have been run



3. Pull Requests

In order to avail better traceability and rigor surrounding the Pull Request process we have developed an in-house solution utilizing Copado. This solution uses a native Salesforce Approval Process to handle all pull requests, while Azure is still used for the source control solution. A webhook connection is used to send PR data (URL, status, etc) from Azure into Copado. Finally, the checkox for "Pull Requests Approved" under the "Definition of Done" criteria on the user story record is updated solely by the PR approval flow. If the PR is approved in Copado the checkbox will be enabled and the user story can promote upward. If the PR is rejected the checkbox will stay disabled until a PR is approved.

- Once a developer has finished building their code and they are ready to promote their User Story to a higher environment they should begin the Pull Request Approval process. To do so follow the steps mentioned in the link - Copado Pull Request Process

(https://humanaprod.service-now.com/kb?id=esc_kb_article&sysparm_article=KB0113335)

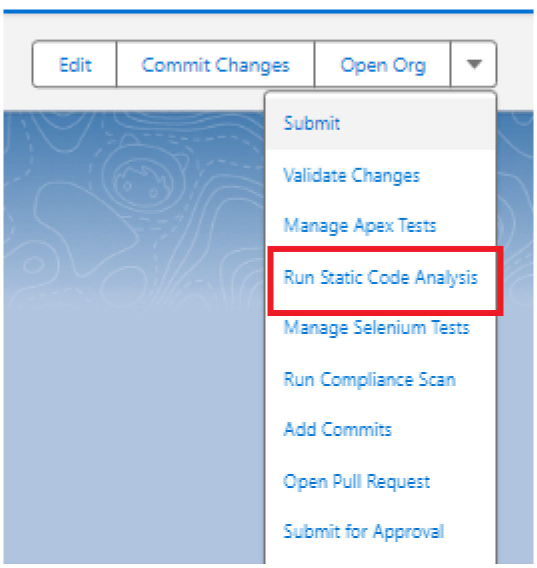
4. Static Code Analysis - PMD Code Scan

Copado includes the static code analysis tool PMD as a built-in feature for performing static code scans on all Apex code for a particular user story. At Humana we utilize a customized ruleset within PMD to detect issues related to security, performance, coding best practices, and more.

All user stories that contain Apex code must be scanned against the PMD ruleset within Copado. A scan has passed if there are no Priority 1 violations found for a given user story. Success or failure of the SCA scan determines the state of the Static Code Analysis Passed checkbox: if the scan passes the checkbox will be enabled, if it fails it will be disabled. This checkbox determines whether your user story can deploy to a higher environment. If a user story contains any Priority 1 violations it will not be allowed to promote and deploy up to SIT. Any attempts to promote and deploy upward will fail, showing the user an error message referencing the Static Code Analysis step.

To run the PMD scan on your User Story please follow the steps below:

- From the **Submit** drop down arrow on the User Story, select **Run Static Code Analysis**



- After a short wait, navigate to the **Related** List items, then **Static Code Analysis Results**
- You will see the output of all rule violations under this section. From here you can see the name of the file that contains the violation, which line it occurred, the rule name, and its priority.
- Finally, you can drill down deeper in the results to get a link directly to the PMD Docs page which includes more info on the rule itself to help you remediate the issue.
- If any Priority 1 violations have been found, fix your code, recommit, and rescan to update the SCA Related List records. If there are no new Priority 1 issues found, the

checkbox for Static Code Analysis Passed will be enabled which will allow you to deploy upward to SIT.

5. SecAPI Scan

- One of the most important checks our Salesforce code goes through prior to Production release is the static Checkmarx scan (Security SAST scan). This scan ensures that code that is being pushed to Production is free of security violations that put our members', business partners', and Humana's own data at risk.
- Refer here for the pre-requistes and process for SecAPI scan (https://humanaprod.service-now.com/kb_view.do?sys_kb_id=e7c0b08e9792e9985727f337f053af30)

6. Back Promotion Checker

- Back Promotion checker checks for pending back promotions count in the target environment and blocks the promotions in target environment,
- Refer here for the Back Promotion Checker (https://humanaprod.service-now.com/kb?id=esc_kb_article&sysparm_article=KB0137979)

Revised by Monisha Rajagopal
Last modified 3 weeks ago

Helpful?

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